

LINUX PROGRAMMING : ASSIGNMENT-1

1. What is Linux Operating System (OS)? List three pros and cons of it.

The Linux OS is a free and open-source operating system built on the Linux kernel. It is a Unix-like operating system that is widely used for servers, desktops, mobile devices, and embedded systems.

ANS Pros of Linux

- * Customization: Linux is highly customizable. You can modify its source code to create an OS that suits your specific needs.

- * Security: Due to its open-source nature and a large community of developers constantly reviewing the code, Linux is considered more secure than many proprietary operating systems.

- * Performance: Linux is known for its stability and reliability, especially in server environments. It can run on a wide range of hardware, from powerful supercomputers to low-power devices.

Cons of Linux

- * Learning Curve: Linux can be challenging for beginners, especially those accustomed to Windows or macOS, and often requires using the command line for many tasks.

- * Software Compatibility: Some proprietary software, particularly for gaming and professional applications, may not be available or fully compatible with Linux.

- * Hardware Support: While improving, some hardware components might not have readily available drivers for Linux, which can lead to compatibility issues.

2. Differentiate between Linux, Mac, Android, and Windows OS with at least six unique features.

ANS • Source Model: Linux and Android are open-source operating systems, which means their source code is freely available to the public. In contrast, Mac OS and Windows are proprietary, or closed-source, systems, with their code owned and controlled by Apple and Microsoft, respectively.

- Primary Use: Linux is widely used for servers, supercomputers, and embedded systems. Mac OS is primarily for desktops and laptops, while Android is the dominant OS for smartphones and tablets. Windows is the most popular desktop OS and is also used for laptops and gaming consoles.

- Kernel Type: Linux uses a monolithic kernel, where all core OS services run in a single address space. Mac OS, Android, and Windows all use a hybrid kernel, which combines aspects of monolithic and microkernels for improved performance and modularity.

- Licensing: Linux is licensed under the GNU General Public License (GPL), which ensures it remains free and open. Android uses licenses like Apache and GPL. Mac OS and Windows have proprietary licenses, and you typically need to purchase them or have them pre-installed on a device.

- **User Interface:** Linux offers a variety of user interfaces through different Desktop Environments (e.g., GNOME, KDE), allowing for extensive customization. Mac OS uses the Aqua interface, Android uses Material Design, and Windows uses Fluent Design.
- **Ecosystem:** The Linux ecosystem is a diverse, community-driven environment with many different distributions. The Mac OS ecosystem is tightly integrated with Apple's hardware and services. Android is part of the Google ecosystem, with a focus on its app store and services. Windows is a central part of the Microsoft ecosystem, including products like Microsoft Office and Azure.

3. Why is Linux preferred for Mainframe Servers for legacy applications? Give three out-of-the-box technical reasons.

Ans. 1. **Stability and Reliability:** Linux has a proven track record of being extremely stable and reliable. This is critical for mainframe environments where applications need to run continuously with minimal downtime. The robust nature of the Linux kernel and its ability to handle high-demand workloads make it a natural fit.

2. **Scalability:** Linux can efficiently handle a wide range of workloads, from a few users to thousands of simultaneous connections. This scalability is essential for mainframes that support large-scale enterprise applications and data processing.

3. **Cost-Effectiveness:** Being open-source, Linux eliminates licensing fees, which can be a significant cost for proprietary OSs. This reduces the total cost of ownership for running legacy applications on mainframe servers.

4. Explain the structure of the Linux File System with proper diagram. Note: you can use the tree command to find it out.

ANS The Linux File System Hierarchy Standard (FHS) is a structured directory tree that organizes all files and directories on the system. It's a single, unified tree starting at the root directory, denoted by /. All other directories and files are branches or leaves of this tree.

* /: The root directory. All other directories and files are located under this directory.

* /bin: Contains essential user binaries (executable files) like ls, cp, and mv.

* /sbin: Contains essential system binaries for system administration, such as halt and reboot.

* /etc: Holds configuration files for system-wide services and applications.

* /home: The directory for user home directories. Each user has their own directory, like /home/user1.

* /lib: Stores essential shared libraries for the binaries in /bin and /sbin.

- * /usr: Contains a hierarchy of read-only user-related data and binaries.
- * /var: Stores variable data, such as log files (/var/log), mail queues, and temporary files (/var/tmp).
- * /dev: Contains device files representing hardware devices.
- * /opt: Used for optional or add-on software from third parties.
- * /tmp: A directory for temporary files created by users and the system, which are often deleted on reboot.

4. If Linux OS is open-source, how do companies like Red Hat still making money from it? Do a market study and answer properly.

ANS While the Linux kernel and many of its components are open-source and free, companies like Red Hat make money by providing commercial services and support. This business model is based on the idea that enterprise customers are willing to pay for reliability, security, and professional assistance, even for free software.

Red Hat's primary source of revenue comes from its subscriptions. These subscriptions provide customers with:

- * Enterprise-Grade Linux Distributions: Red Hat takes the open-source Linux kernel and packages it into a stable, secure, and well-tested enterprise distribution, such as Red Hat Enterprise Linux (RHEL).
- * Technical Support: Customers pay for access to a team of experts who can help troubleshoot issues, provide guidance, and ensure their systems are running smoothly. This is crucial for businesses that need guaranteed uptime.
- * Security and Maintenance Updates: Red Hat provides regular security patches, bug fixes, and maintenance updates to protect systems from vulnerabilities. This is a significant value proposition for businesses concerned about security.
- * Consulting and Training: Red Hat also offers consulting services to help companies implement and optimize their Linux infrastructure, as well as training to get their IT staff up to speed.

This model allows Red Hat to monetize the open-source software by selling the services that wrap around it, addressing the needs of enterprise clients who require more than just the software itself.

6. Write the command to display today's date and time (i.e., current System time).

ANS The command to display the current date and time in Linux is date

7. Which command is used to check how long the system has been running?

ANS The command to check how long the system has been running is uptime.

8. What is the difference between shutdown -h now and halt?

ANS Both shutdown -h now and halt are used to stop a Linux system, but they differ in how they do it.

- * shutdown -h now: This is a more graceful and recommended way to shut down. The command sends a signal to all processes to terminate, unmounts file systems, and then powers off the system. It's considered safer as it ensures all data is saved and processes are closed properly before the machine is powered down.

- * halt: This command is a lower-level command that immediately stops the system and shuts down the CPU. It doesn't perform the graceful shutdown procedures that shutdown does. While it can be faster, it may not close all processes or unmount all file systems properly, which could lead to data corruption or other issues

9. Compare init 0 and shutdown -h. Which is safer? Why?

ANS Shutdown -h is safer because it is designed to perform a controlled and graceful shutdown. It first sends signals to all running processes to terminate, unmounts the file systems, and then powers down the system. This process helps prevent data loss or corruption, which can occur with a less-controlled command like init 0.

- * Primary Action: init 0 changes the system's runlevel to 0, which is the designated runlevel for halting the system. shutdown -h explicitly powers down (halts) the system.

- * Graceful Shutdown: shutdown -h is designed to be more graceful and controlled; it initiates an orderly sequence of stopping services, notifying users, and unmounting filesystems. init 0 changes the runlevel, which is a less graceful method that may not notify all users or properly close all applications.

- * User Notification: shutdown -h can broadcast a shutdown message to all logged-in users, giving them a chance to save their work and log out. init 0 does not have a built-in feature for broadcasting a shutdown message.

- * Safety Implication: shutdown -h is considered the safer command. init 0 is less safe, as it might not properly close all applications or save data before halting the system

10. A system administrator accidentally powers off a Server machine without shutting it down properly. What problems can occur to the said Server?

ANS Powering off a server without a proper shutdown can cause several problems:

- * File System Corruption: When a system is powered off abruptly, files may not be properly written to the disk, leading to an inconsistent state. This can corrupt the file system, making it difficult or impossible to boot the system again without using recovery tools.

- * Data Loss or Corruption: Any data that was in the process of being written to the disk at the time of the power-off may be lost or corrupted. This is especially risky for databases and other applications that require data integrity.

- * Unclean Shutdown Issues: Applications and services may not have a chance to properly close their files and processes. This can leave behind locked files, orphaned processes, and other inconsistencies that require manual intervention to fix.

Brainstorming

a) As Linux Kernel is open-source, can we build our own operating system?

since the Linux kernel is open-source, you can use it to build your own operating system. Many operating systems, such as Android, Chrome OS, and various Linux distributions like Ubuntu and Fedora, are based on the Linux kernel

b) In order to do that, what are the stoppers, hurdles, and challenges?

Building a new operating system from the Linux kernel involves significant challenges:

- * Integration and Development: You'd need to integrate a userland—the collection of programs, libraries, and utilities that make the OS functional—with the kernel. This includes choosing or creating a shell, a desktop environment (like GNOME or KDE), and a package management system.
- * Hardware Compatibility: Creating an OS that works with a wide range of hardware requires a deep understanding of drivers and hardware architecture.
- * Security and Maintenance: Maintaining an OS involves a continuous effort to provide security patches, bug fixes, and updates. This requires a dedicated team and robust infrastructure.

c) Is anyone in India working on this field? Find at least three to four engineers.

While it's difficult to provide an exhaustive list, there are many engineers in India who contribute to the Linux kernel and related projects. Some notable individuals include:

- * Amit Singh (contributor to the Linux kernel and other open-source projects).
 - * Srivatsa Vaddagiri (a Linux kernel developer known for his work on scheduler and power management).
 - * Arjun Shankar (involved in various open-source projects and Linux kernel development).
- These individuals, among others, contribute to the global open-source community, helping to develop and improve the Linux kernel.